FORECASTING: PART 1

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What is Forecasting?

- Forecasting refers to **the practice of predicting what will happen in the future by taking into consideration events in the past and present.**
- Forecasting is a decision-making tool that helps businesses cope with the impact of the future's uncertainty by examining historical data and trends.

- In Supply Chain, forecasting refers to the process of predicting **demand, supply or pricing for a product** — or a range of products — in a particular industry.

https://www.gep.com/knowledge-bank/glossary/what-is-supply-chain-forecasting

Forecasting Time horizons

- **Short range forecast**
  - This forecast is usually used for three months or less. It can be up to one year,
  - It is used for planning purchases, hiring, job assignments, production levels, and the like.
- **Medium range forecast**
  - This is for three months to three years.
  - Medium range forecasts are used for sales and production planning, budgeting, and analysis of different operating plans.
- **Long range forecast**
  - Three years or more in time span.
  - Used for new products, capital expenditures, facility expansion, relocation, and research and development.

https://www.universalclass.com/articles/business/the-art-and-science-of-forecasting-in-operations-management.htm

Medium and long range forecasts

- More comprehensive in nature
- Support and guide management decisions in planning products, processes, and plants.

Short term forecasts

- To anticipate immediate future needs and impacts, short term forecasts use existing data in mathematical formulas.
- Quantitative in nature and use.
- Relatively more accurate than medium or long range forecasts.
- Short term forecasts are to be updated regularly to maintain their effectiveness.

Types of Forecasting

- **Economic forecasts**
  - **It** addresses the business cycle. They predict inflation rates, money supplies, and other indicators.
- **Technological forecasts**
  - **It** monitors the rates of technological progress. This keeps organizations aware of trends and can result in exciting new products.
  - New products may require new facilities and equipment. It must be planned for in the appropriate time frame.
- **Demand forecasts**
  - It deals with the company's products and estimates consumer demand.
  - Also referred to as sales forecasts. Have multiple purposes like driving scheduling, production, and capacity.

https://www.universalclass.com/articles/business/the-art-and-science-of-forecasting-in-operations-management.htm

Steps in Forecasting System

- Determine what the forecast is for (e.g., rainfall, cyclone, market trend, demand of a product)
- Select the items to be forecasted
- Select the time horizon (Is it short, medium, or long term?)
- Select the forecast model type (quantitative, qualitative)
- Gather data to be input into the model
- Make the forecast
- Verify and implement the results

Forecasting Approaches

- Two general approaches of forecasting are **qualitative approach** and quantitative analysis.
- Qualitative forecasts are a subjective approach. It incorporates factors such as the decision maker's intuition, emotions, personal experiences, and value system in reaching a forecast. Some firms use one approach and some use the other.
- Quantitative forecasts use different mathematical models. These rely on historical data and their associative variables to forecast demand.

Qualitative forecasting techniques

- **Jury of executive opinion**
  - It is based on the inputs and decisions of high-level experts or management.
- **Delphi method**
  - Decision makers, staff, and respondents all meet to develop the forecast. Every shareholder in the process provides input. Reach a group consensus on a complex issue by gathering opinions from a panel of experts in a series of rounds.
- **Sales force composite**
  - Each sales person provides an individual estimate which is reviewed for realism by management, and then combined for a big picture view.
- **Consumer market survey.**
  - This is surveying the prospective customer base to determine demand for existing products and can also be used for new products.

Quantitative Analysis

- **Extrapolative or time series methods**
  - It uses past history of demand in making a forecast for the future
  - The pattern in historical data is identified and extrapolation is done
  - For shorter time horizons, this method works well
- **Casual or explanatory methods**
  - It assumes that the demand for an item depends on one or more independent factors (e.g., price, advertising)
  - A relationship between different variables is established.

Extrapolative/Time series Methods

- Extrapolative methods identify patterns in past data
- These patterns depend on four components of demand:
  - **horizontal, trend, seasonal, and cyclical**.
- The horizontal component of demand exists when the demand fluctuates about an average demand.
- The trend component of demand refers to a sustained increase or decrease in demand from one period to the next.
- The seasonal component of demand pertains to the influence of seasonal factors that impact demand positively or negatively.
- The cyclical component of demand is similar to the seasonal component except that seasonality occurs at regular intervals and is of constant length.

Moving Average Method

- It is the simplest extrapolative method. Two steps are required to make a forecast for the next period from past data.

  1. Select the number of periods for which moving averages will be computed. This number, *N*, is called an order of moving average.
  2. Take the average demand for the most recent *N* periods. This average demand then becomes the forecast for the next period.

Examples of Moving Average Method

- **Example**: The demand data for a product in the months of January, February, March, April and May is 100, 90, 120, 80 and 110 units, respectively. Forecast the demand for the month of June.
- **Solution:**

  Choose *N* = 4 (say). Other values can also be chosen. Larger *N* values will have a greater smoothing effect on random fluctuations in demand.

  The average demand for the most recent 4 periods, *N* = 4 will be estimated.

$$\text{Moving average} = \frac{\text{Demand for February, March, April, May}}{4}$$

$$= \frac{90+120+80+110}{4} = 100 \text{ units}$$

The forecast for June is **100 units**.

Examples of Moving Average Method

- Now, suppose the actual demand for June turns out to be 90 units. The forecast for July will be computed by taking the average demand for March, April, May and June. The forecast of July will be

$$\frac{120+80+110+90}{4} = 100 \text{ units}$$

- After *N* is selected, the new moving average for each future period is computed by taking the average demand for the most recent *N* periods.
- The disadvantage of this method is that it requires the storage of demand data for *N* periods for each item
- Also, this method is not applicable if the demand data reflect trends or seasonal components. If there is an upward trend, the forecast made by this method will underestimate the actual demand.

Exponential Smoothing Method

- In this method, the weight assigned to a previous period's demand decreases exponentially as that data gets older.
- The recent demand data receives a higher weight than older demand data.
- Exponential smoothing methods are particularly attractive for production and operations applications that involve forecasting for a large number of items.
- The forecasting horizon is relatively short (daily, weekly, or monthly demand needs to be forecasted)
- This method is used when there is little "outside" information available about cause and effect relationships between the demand for an item and independent factors that influence it.

Basic Exponential Smoothing Model

- The simplest exponential smoothing model is applicable when there is no trend or seasonality component in the data.
- Only the horizontal component of demand is present. Because of randomness, the demand fluctuates around an "average demand," which is called as the "base."
- If the base is constant from period to period, then all fluctuations in demand must be due to randomness.
- In reality, fluctuations in demand are caused by both changes in the base and random noise.

- **The key objective in exponential smoothing models is to estimate the base and use that estimate for forecasting future demand.**

Basic Exponential Smoothing Method

- In the basic exponential smoothing model, the base for the current period, S_t, is estimated as

New base = Previous base + α(Actual current demand – Previous base)

- In mathematical symbols

$$S_t = S_{t-1} + \alpha\left(D_t - S_{t-1}\right) \qquad (1)$$

S_t = Base for the current period, S_{t-1} = Previous base, D_t = Actual current demand

α (alpha) = smoothing constant, that lies between 0 and 1

❑ $D_t - S_{t-1}$ is the observed error e for period t

Basic Exponential Smoothing Method

- Equation 1 can also be expressed as

$$S_t = \alpha D_t + (1-\alpha) S_{t-1}$$

Properties of the Exponential Smoothing Model

- This method is simple and requires that minimal information be stored for each item.
- Unlike the moving average method, for which all past *N* observations must be stored, the exponential smoothing model requires only two pieces of data: the most recent demand and the previous base.
- The procedure can be easily automated, and the data entry in each period is only the new demand information.

Basic Exponential Smoothing Method

- **Example:** A mobile phone manufacturer forecasted the sales of mobile phones in February to be 140 units. However, the actual sale was 150 units. By exponential smoothing method with a smoothing constant of α = 0.2, the sales forecast for the month of March is _____ units.
- **Solution**

Given, S_{t-1} = Forecast for February = 140 units
D_t = Actual current demand for February = 150 units
α (alpha) = 0.2

To find,
S_t = Forecast for the month of March

$$S_t = \alpha D_t + (1-\alpha) S_{t-1}$$
$$= (0.2 \times 150) + (0.8 \times 140) = 142 \text{ units}$$

Properties of the Exponential Smoothing Model

- Small values of α (alpha) will have a stronger smoothing effect than large values.
- Conversely, large values of α will reflect real changes in actual demand (as well as random variations) more quickly.
- If fluctuations in demand are primarily due to randomness, a small α should be chosen.
- If fluctuations in demand are due to a shifting base, then a higher α should be chosen.

Properties of the Exponential Smoothing Model

$$S_t = \alpha D_t + (1-\alpha) S_{t-1}$$

- In extreme cases, the values are $\alpha = 0$ and $\alpha = 1$

- If $\alpha = 0$, then $S_t = S_{t-1}$
  - In this case, the new base is the same as the previous base. All fluctuations in demand are attributed to randomness

- If $\alpha = 1$, then $S_t = D_t$
  - In this case, new base is the new demand and the previous base is totally ignored.
  - It is assumed that the new demand has no randomness and the base has in fact shifted.

Properties of the Exponential Smoothing Model

- A small α (alpha) has a greater smoothing effect on demand. It is usually recommended that an α between 0.01 and 0.30 be chosen.
- It is also possible to conduct an empirical test in which *a* (alpha) is varied and forecast errors are computed for several alternative values of α (alpha). The value of α (alpha) that produces minimum forecast error is then chosen.

It is important to recognize that at least one year of historical data are required before confidence can be placed in the resulting forecasts.

FORECAST ERRORS

- Forecast error (e_t) is defined as

  e_t = Actual demand for period t — Forecast for period t

- Average error (A_E)= $\frac{1}{N}\sum_{t=1}^{N} e_t$

  - The average error (A_E) should be near zero for a larger sample; otherwise, the model exhibits bias. Bias indicates a systematic tendency to forecast high or low values for demand. However, A_E obscures variability because positive and negative errors cancel out.

FORECAST ERRORS

- **Mean absolute deviation (MAD) =** $\frac{1}{N}\sum_{t=1}^{N}|e_t|$

- The mean absolute deviation (MAD) provides additional information that is useful in selecting a forecasting model and its parameters.
- MAD is simply the sum of all errors without regard to algebraic sign, divided by the number of observations

FORECAST ERRORS

- **Mean squared error(MSE) =** $\frac{1}{N}\sum_{t=1}^{N} e_t^2$

- The mean squared error (MSE) provides information similar to MAD, but it penalizes larger errors. MSE is computed by summing the squared individual errors and dividing by the number of observations

FORECAST ERRORS

- **Mean absolute percentage error (MAPE) =** $$\frac{1}{N}\sum_{t=1}^{N}\left|\frac{e_t}{D_t}\times 100\right|$$

- The mean absolute percentage error (MAPE) is a relative measure that is computed by dividing the forecast error for period *t* by the actual demand for period *t* and thereby computing the percentage error in period *t*.

Example

- Calculate the forecast error by all four methods.

Period	Demand	Forecast
1	20	22
2	40	38
3	10	13
4	20	16
5	30	32

Forecast error *e* for period *t* = $D_t - S_{t-1}$

$$\%\ \text{error in period } t = \frac{\text{Forecast error in period } t}{\text{Demand in period } t} \times 100$$

Example

Period	Demand	Forecast
1	20	22
2	40	37
3	10	13
4	20	16
5	30	32

Error	$(\text{Error})^2$	% error
–2	4	**–10%**
3	9	7.5%
7	49	70%
4	16	20%
–2	4	**–6.67%**

$$\text{Average error} = \frac{-2+3+7+4-2}{5} = 2$$

$$\text{Mean squared error} = \frac{4+9+49+16+4}{5} = 16.4$$

$$\text{Mean absolute deviation} = \frac{2+3+7+4+2}{5} = 3.6$$

$$\text{Mean absolute percentage error} = \frac{10+7.5+70+20+46.67}{5} = 30.84$$

Classification of Exponential Smoothing Models

- LINEAR OR ADDITIVE TREND MODEL
- RATIO SEASONALITY MODEL
- ADDITIVE TREND AND RATIO SEASONALITY MODEL
- INITIALIZATION
- ADAPTIVE METHODS

Linear or Additive Trend Model

- The apparent linear trend in exponentially smoothed averages is the difference between the successive values, $S_t - S_{t-1}$
- Sometimes a negative trend is recorded when, in fact, the general trend is positive. To minimize these irregular effects, we stabilize the raw trend measure in the same way we stabilized actual demands, by applying exponential smoothing.
- **$S_t - S_{t-1}$ series can be smoothened with the smoothing constant β (beta).**
- This smoothing constant need not have the same value as the α used in smoothing D_t.

Linear or Additive Trend Model

- The mathematical expression of basic exponential smoothing model can be modified by simply adding the old smoothed trend to the previous base as follows:

$$S_t = \alpha D_t + (1-\alpha)(S_{t-1} + T_{t-1}) \qquad \text{(A)}$$

T_{t-1} = Trend estimate last period

- The updated value of T_t, the smoothed trend is

$$T_t = \beta(S_t - S_{t-1}) + (1-\beta)T_{t-1} \qquad \text{(B)}$$

Linear or Additive Trend Model

- Equations A and B yield smoothed current values of demand and trend.
- Therefore to forecast the upcoming period, we extrapolate by adding T_t to the current smoothed average, S_t, as follows:

$$\text{Forecast for the next period} = F_{t,1} = S_t + T_t$$

$$\text{Forecast for } m \text{ periods ahead} = F_{t,m} = S_t + mT_t$$

This concept is useful in some of the more complex models that combine estimates of average demand, trend, and seasonal components.

Example

- A manufacturer wants to forecast demand for a piece of pollution-control equipment. A review of past sales is shown. The firm assumes the initial forecast average for month 1 was 11 units and the trend over that period was 2 units. Smoothing constants are $\alpha = 0.2$, $\beta = 0.4$.

Month	Actual Demand
1	13
2	17
3	20

- Solution

Forecast average for Month 2

$$S_t = \alpha D_t + (1-\alpha)(S_{t-1} + T_{t-1})$$
$$= (0.2 \times 13) + (1-0.2) \times (11+2)$$
$$= 13 \text{ units}$$

Example

- Trend in period 2

$$T_t = \beta(S_t - S_{t-1}) + (1-\beta)T_{t-1}$$
$$= 0.4 \times (13-11) + (1-0.4)2$$
$$= 2$$

- Forecast including trend = $S_t + T_t$

= 13 + 2 = 15 units